

# Communication-Efficient Federated Learning for Post-Flood Risk Assessment Using UAV Swarms

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## Abstract

Accurate semantic segmentation is crucial for rapid risk and severity assessment following flash flood disasters. However, in drone networks deployed for disaster recovery, bandwidth constraints mean the backhaul transmission of massive imagery required for centralized processing is highly prone to causing network congestion. Federated Learning, a decentralized collaborative paradigm that permits drones to process data locally, offers an ideal pathway to address this challenge. Nonetheless, existing FL methods are plagued by the dilemma of excessive communication overhead and insufficient segmentation performance. Therefore, this paper proposes a Communication-Efficient Federated Distillation (CEFD) framework. The core of this framework lies in the design of a lightweight, multi-level knowledge representation. It discards the transmission of bulky parameters, opting instead for the efficient exchange of high-level semantic logits and key intermediate features that have undergone feature-space dimensionality reduction. On the server side, adaptive weighted aggregation is then utilized to construct a robust global knowledge model. Experimental results show that CEFD achieved high segmentation performance on the LSD dataset, attaining an mIoU of 0.6226, a Mean Recall of 0.6532, an Accuracy of 0.9794, and an MAE of 0.0400, while reducing communication overhead by approximately 99.5%. This enables scalable collaborative intelligence for resource-constrained edge devices, a key capability for the future web systems.

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## CCS Concepts

• **Computing methodologies** → **Distributed artificial intelligence**; • **Networks** → *Network resources allocation*; *Mobile networks*.

## Keywords

Post-disaster assessment; Federated learning; Efficient communication; Image segmentation; UAV Swarms network

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## 1 Introduction

Sudden disasters such as landslides and debris flows pose a severe threat to human life and property [27, 29]. In the aftermath of a disaster, rapid assessment of risk and severity is the primary task for emergency response [17]. This assessment relies on the precise semantic segmentation of flash flood inundation zones and landslide boundaries to support rescue decisions. However, these disasters often destroy ground-based communication infrastructure, turning the affected area into an information island and impeding any effective data collection. Against this backdrop, aerial ad-hoc networks composed of Unmanned Aerial Vehicle (UAV) swarms have emerged as a critical disaster assessment network technology [14, 20]. They not only serve as temporary communication relays but can also collaboratively enter areas inaccessible to human personnel to collect the high-definition imagery required for assessment, enabling the possibility of real-time disaster severity evaluation [13].

To achieve rapid risk and severity assessment in post-flash-flood recovery scenarios, Unmanned Aerial Vehicle (UAV) networks must

possess near real-time semantic segmentation capabilities to automatically identify and map flood inundation zones and landslide boundaries [2, 30]. However, a fundamental contradiction exists between the data throughput required to run these models and the inherently fragile bandwidth of aerial UAV networks. Adopting a centralized processing approach, which involves the real-time backhaul of high-definition video streams from all UAVs, would immediately lead to network congestion [4]. Federated Learning (FL) offers a data-decentralized collaborative training approach [1, 6, 31], which allows UAVs to process data locally, thereby avoiding the transmission of raw imagery. However, traditional FL was not originally designed for disaster recovery networks. It requires the frequent exchange of complete model parameters, often hundreds of megabytes in size, across the network [5, 22, 24]. This high communication cost, for bandwidth-constrained and unstably connected UAV networks, would induce catastrophic latency and extremely high transmission failure rates, rendering it impractical for application [26]. To specifically address the high communication costs of traditional FL, an alternative called the federated knowledge distillation method has been proposed. This method effectively minimizes network load by exchanging only lightweight logits but introduces severe performance pitfalls. Such an extreme pursuit of efficiency sacrifices information integrity, resulting in the loss of critical low-level spatial structural details. Consequently, the model is rendered incapable of precisely delineating disaster boundaries, failing to meet the stringent standards required for emergency response. Therefore existing technologies are trapped in a dilemma between communication overhead and segmentation performance.

This paper proposes a Communication-Efficient Federated Distillation (CEFD) framework characterized by the decoupling and selective transfer of logits and compressed intermediate features. By leveraging these complementary knowledge sources, the framework balances low communication costs with high segmentation accuracy. This lightweight yet information-rich mechanism facilitates precise segmentation in bandwidth-constrained networks, thereby fundamentally resolving the aforementioned dilemma.

The main contributions of this paper are as follows:

- (1) We propose the CEFD framework, which introduces a novel multi-level knowledge representation to decouple and selectively transfer high-level semantic knowledge and essential spatial structure knowledge, addressing the accuracy and efficiency tradeoff in distributed learning at the network edge.
- (2) We designed a Feature Spatial Compression (FSC) mechanism that serves not only as a network compression tool but also as an information refinement method, enhancing the robustness of aggregated knowledge on Non-Independent and Identically Distributed (Non-IID) data.
- (3) Experimental results demonstrate that CEFD achieves high segmentation performance on the LSD dataset with an mIoU of 0.6226, a Mean Recall of 0.6532, an Accuracy of 0.9794, and an MAE of 0.0400, while reducing communication overhead by approximately 99.5% compared to parameter aggregation methods. The significant improvement in communication

efficiency not only makes it suitable for Web-scale data generation devices but also directly reduces energy consumption, which is critical for battery-powered edge devices.

The remainder of this paper is organized as follows. Section 2 reviews the related work. Section 3 describes the proposed methodology. Experiments and analysis are presented in Section 4, and conclusion are drawn in Section 5.

## 2 Related Work

### 2.1 Deep Learning Segmentation Techniques for Disaster Assessment

In recent years, researchers have made significant progress in utilizing deep learning segmentation techniques to enhance the accuracy and robustness of automated disaster assessment. A primary challenge in disaster assessment stems from the complexity of high-resolution imagery. To address this, Diakogiannis et al. [3] proposed the ResUNet-a framework, which integrates residual connections, atrous convolutions, and pyramid pooling to tackle pixel-level variance challenges in high-resolution images. Beyond pixel value variance, the confusability of disaster features is another major difficulty. Qurratulain et al. [23] utilized a ResNet instance segmentation model to more accurately distinguish between easily confused features, such as burned areas and bare ground. Building upon innovations in model architecture, fusing multi-source data is another effective pathway to improve assessment accuracy. For example, Lu et al. [18] designed a dual-encoder architecture to enhance landslide detection performance by fusing deep features from optical imagery and DEM data. Following a similar multi-modal fusion approach, Koppika et al. [11] also adopted a multi-scale method, combining SAR image analysis with deep learning segmentation to achieve multi-level damage detection for infrastructure. Furthermore, targeting road network assessment, which is critical in post-disaster recovery, Gupta et al. [7] proposed a framework utilizing OSM data and graph theory to focus on identifying passable road networks post-disaster.

All the advanced segmentation models mentioned above rely on a centralized training paradigm. However, this prerequisite is untenable in our post-flash-flood recovery scenario. Because the disaster area data is inherently collected in a distributed manner by multiple UAVs and the ground network has been destroyed, exploring a decentralized collaborative training framework that does not require sharing raw data becomes an inevitable choice.

### 2.2 Communication-Efficient Federated Learning for Segmentation Tasks

FL, as a decentralized collaborative paradigm, presents an ideal pathway for addressing the collaborative assessment challenges in UAV swarms. However, standard FL algorithms, such as FedAvg, require the frequent transmission of bulky model parameters within the disaster recovery network, which introduces substantial communication overhead. Consequently, researchers have explored various communication-efficient FL strategies. For instance, to solve the problems of large storage and communication overhead in FL, Shah et al. [25] proposed a model compression scheme. This scheme designs separate compression strategies for the server and clients to

collaboratively build and evolve sparse models, ultimately outperforming existing state-of-the-art baselines in both model accuracy and communication efficiency. Targeting the challenges of communication constraints and data heterogeneity, Hu et al. [8] proposed a communication-efficient personalized FL framework. It employs probabilistic mask optimization in place of weight training to generate personalized sparse sub-networks and combines this with a binarization communication strategy. While significantly reducing communication overhead, its model accuracy also surpasses existing advanced methods. Furthermore, Kouda et al. [12] addressed communication constraints in wireless IoT environments by proposing a novel algorithm named Communication-Aware Federated Averaging (CAFA). This algorithm leverages the idle computational capacity during the communication phase to perform local training and model aggregation. Experimental results show that CAFA can dramatically reduce communication costs while maintaining the same level of model accuracy. Liang et al. [16], focusing on the unique feature distribution shift problem in federated semantic segmentation, proposed a communication-efficient adversarial contrastive FL method. This method uses a global discriminator and prototype contrastive learning for adversarial training to learn domain-invariant features, and utilizes a Gaussian Mixture Model to sample virtual features to enhance communication efficiency, demonstrating its superiority on multiple datasets.

Previous approaches have improved communication efficiency using model compression and sparsification but typically overlook the low-level spatial structural information, such as boundaries and textures, required for high-precision segmentation in disaster assessment. The resulting loss of key details during compression renders these methods unable to reconcile the trade-off between high precision and high efficiency in disaster assessment contexts.

### 3 Methodology

To achieve rapid, precise situational awareness within the severe constraints of post-flash-flood recovery, we propose CEFD framework. This framework is specifically designed for UAV swarm networks operating under limited communication bandwidth and with highly heterogeneous Non-IID data. Its core design abandons the high-cost parameter transmission of traditional FL, instead adopting a multi-level knowledge transfer mechanism. This approach maintains high segmentation accuracy while minimizing network communication overhead.

#### 3.1 Algorithm framework

The CEFD framework operates within a server-client topology, wherein the ground command center functions as the Server ( $S$ ), and each UAV serves as a Client ( $c$ ). During the  $t$ -th training round, each client  $c$  trains its local segmentation model using its locally acquired disaster imagery data. Following training, the client does not upload the bulky model parameters. Instead, it extracts two crucial forms of knowledge. The first is high-level semantic knowledge, specifically the model's final output logits, denoted as  $z_c$ . The second is mid-level structural knowledge: the feature map  $\phi_c^{(l)}$  extracted from an intermediate model layer  $l$ , which encapsulates rich spatial and textual information. Based on our DeepLabV3+ architecture we selected four layers spanning the encoder and decoder

as  $\phi_c^{(l)}$  to capture a complete feature spectrum from fine-grained textures to abstract semantics. On the encoder side, it includes low-level features, which provide fine-grained texture and edge information, and aspp features, which capture multi-scale semantic context. On the decoder side, it includes shortcut features, which are responsible for reinjecting shallow high-resolution spatial details into the decoder path, and the final fused features that fuse semantics and space. This multi-level combination is paramount for the precise delineation of complex boundaries for flash floods and landslides.

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#### Algorithm 1 The CEFD Framework

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**Input:** Total communication rounds  $R$ , number of clients  $N$ , local epochs  $E$ , server  $S$

**Server Executes:**

Initialize global knowledge  $\{z_S, \psi_S^{(l)}\}$

**for** each round  $t = 1, 2, \dots, R$  **do**

$C_t \leftarrow$  (Select a subset of clients)

**for** each client  $c \in C_t$  **in parallel do**

$\{z_c, \phi_c^{(l)}\} \leftarrow \text{ClientUpdate}(c, \{z_S, \psi_S^{(l)}\})$

**end for**

{Server Aggregation}

Receive  $\{z_c, \psi_c^{(l)}\}$  from all clients in  $C_t$

Compute  $Q_c$  for all  $c \in C_t$

Aggregate  $\{z_S, \psi_S^{(l)}\}$  using  $\{z_c, \psi_c^{(l)}\}$  and  $Q_c$

Broadcast  $\{z_S, \psi_S^{(l)}\}$  to all clients

**end for**

**ClientUpdate**( $c, \{z_S, \psi_S^{(l)}\}$ )

{Client Local Update}

Set received global knowledge  $\{z_S, \psi_S^{(l)}\}$

**for** each local epoch  $e = 1 \dots E$  **do**

Compute loss  $\mathcal{L}_{node} = \mathcal{L}_{seg} + \gamma(\mathcal{L}_{out} + \mathcal{L}_{feat})$

Update local model via gradient descent

**end for**

Extract logits  $z_c$  and features  $\phi_c^{(l)}$

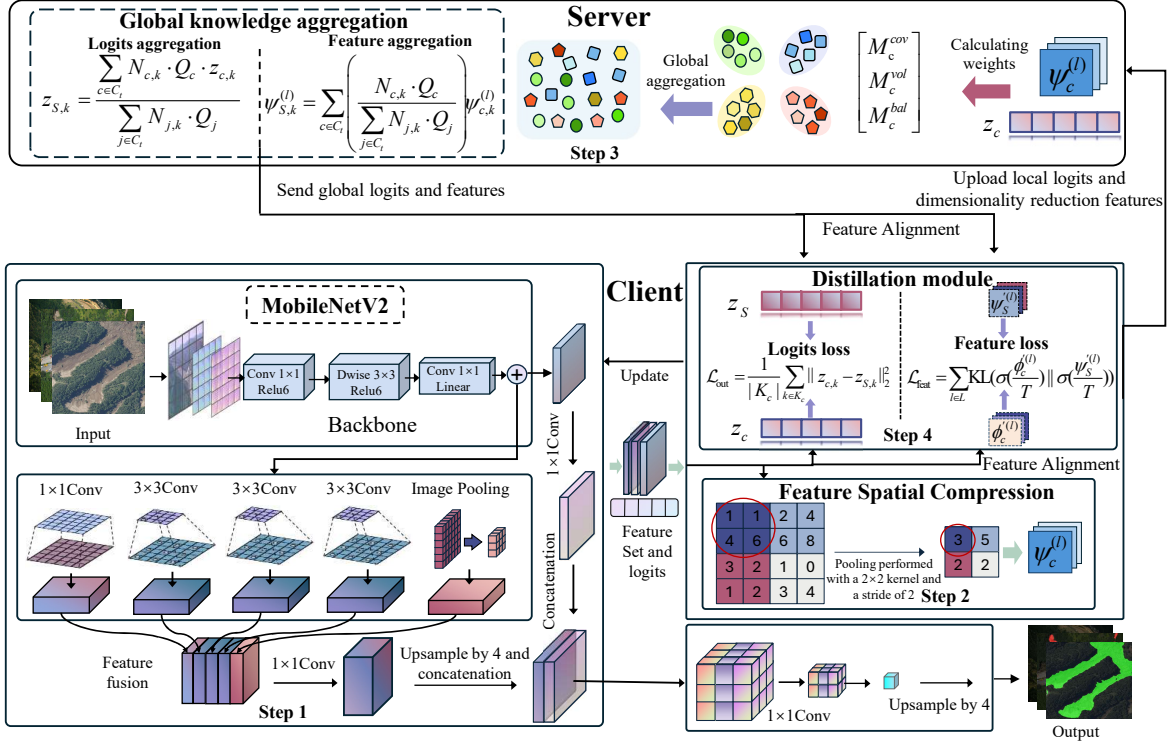
Compress features:  $\psi_c^{(l)} \leftarrow \text{FSC}(\phi_c^{(l)})$

**return**  $\{z_c, \psi_c^{(l)}\}$  to Server  $S$

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To ensure network-efficient transmission, the mid-level feature map  $\phi_c^{(l)}$  undergoes processing by a FSC module (such as adaptive pooling), compressing it into a low-dimensional yet information-dense representation,  $\psi_c^{(l)}$ . Subsequently, client  $c$  uploads this lightweight knowledge pair  $\{z_c, \psi_c^{(l)}\}$  to the server.

The server  $S$  gathers the knowledge packages  $\{z_c, \psi_c^{(l)}\}$  from all UAV clients. It first computes a dynamic contribution factor  $Q_c$  for each client based on their data quality, which includes metrics such as class coverage, data volume, and equilibrium. The server then performs a weighted aggregation of the collected knowledge to generate a more comprehensive and robust global knowledge model, comprising global logits  $z_S$  and global features  $\psi_S^{(l)}$ . Finally, the server disseminates this refined global knowledge  $\{z_S, \psi_S^{(l)}\}$  back to all UAV clients. Clients utilize this as a "teacher" signal, guiding the next round of updates for their local models through a multi-level knowledge distillation loss. The overall architecture



**Figure 1: The overall architecture of the proposed CEFD framework. Samples first pass through the Step 1 model module to extract logits and intermediate-layer features. The extracted intermediate-layer features are then processed by the Step 2 feature spatial compression for dimensionality reduction. These reduced-dimension features, along with the logits, are uploaded to the Step 3 server module for aggregation. Finally, the aggregated knowledge is distributed to the clients and passed through the Step 4 distillation module for model updating.**

of the framework is illustrated in Fig. 1 and its detailed execution process is described in Algorithm 1.

### 3.2 Disaster-Aware Multi-Level Distillation

In disaster recovery scenarios, the high degree of data heterogeneity presents a significant challenge. To address this issue, the local training for each client  $c$  employs a composite loss function  $\mathcal{L}_{\text{node}}$  as defined below:

$$\mathcal{L}_{\text{node}} = \mathcal{L}_{\text{seg}} + \gamma(\mathcal{L}_{\text{out}} + \mathcal{L}_{\text{feat}}) \quad (1)$$

This loss comprises the standard local segmentation loss  $\mathcal{L}_{\text{seg}}$ , and a multi-level distillation loss, composed of  $\mathcal{L}_{\text{out}}$  and  $\mathcal{L}_{\text{feat}}$ , used for aligning with the global knowledge.  $\gamma$  is a hyperparameter that balances these two loss components.

$\mathcal{L}_{\text{seg}}$  is the standard supervised learning loss on the local data  $D_c$ . We adopt the Weighted Cross-Entropy loss to handle the data imbalance among disaster categories, such as landslide, debris flow, and background. It is defined as follows:

$$\mathcal{L}_{\text{seg}} = -\frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} \sum_{k=1}^{|K|} \omega_k \cdot y_{i,k} \cdot \log(\sigma(z_{c,i,k})) \quad (2)$$

where  $\mathcal{P}$  is the set of pixels,  $|K|$  is the total number of classes,  $\omega_k$  is the weight for class  $k$ ,  $y_{i,k}$  is the ground-truth label for pixel  $i$ ,

$\sigma(\cdot)$  is the Softmax function, and  $z_{c,i,k}$  is the local model's predicted logit for pixel  $i$  belonging to class  $k$ .

$\mathcal{L}_{\text{out}}$  represents the output-layer distillation loss, which guides the local model's final output  $z_c$  to mimic the high-level semantic distribution within the global knowledge  $z_S$ . We use the  $L_2$  norm (Mean Squared Error) to measure the average difference between them for class  $k$ , which guides the local model to learn the global consensus.

$$\mathcal{L}_{\text{out}} = \frac{1}{|K_c|} \sum_{k \in K_c} \|z_{c,k} - z_{S,k}\|_2^2 \quad (3)$$

where  $K_c$  is the set of classes possessed by client  $c$ ,  $z_{c,k}$  are the logits predicted by the local model of client  $c$  for class  $k$ , and  $z_{S,k}$  are the corresponding global logits for class  $k$  aggregated by the server.

$\mathcal{L}_{\text{feat}}$  is the feature-layer distillation loss, used to transfer finer-grained disaster textures and spatial structure information. Due to potential dimensional disparities between the local features  $\phi_c^{(l)}$  of different clients and the server-distributed global features  $\psi_S^{(l)}$  we first introduce a lightweight feature alignment network  $A_{\text{align}}$ .  $A_{\text{align}}$  is a lightweight  $1 \times 1$  convolutional block projecting the features into a unified space while minimizing overhead. It maps both to a unified dimensional space yielding the aligned features  $\phi_c'^{(l)}$  and  $\psi_S'^{(l)}$ :

$$\phi_c'^{(l)} = A_{\text{align}}(\phi_c^{(l)}) \quad (4)$$

$$\psi_S'^{(l)} = A_{align}(\psi_S^{(l)}) \quad (5)$$

To make the aligned local features  $\phi_c'^{(l)}$  approximate the aligned global features  $\psi_S'^{(l)}$ , we employ the KL divergence paired with a temperature coefficient  $T$  to soften the distributions, thereby achieving a smoother knowledge transfer.

$$\mathcal{L}_{feat} = \sum_{l \in L} \text{KL}(\sigma(\phi_c'^{(l)}/T) || \sigma(\psi_S'^{(l)}/T)) \quad (6)$$

where  $L$  is the set of selected intermediate layers,  $\sigma(\cdot)$  represents the Softmax function, which is applied along the spatial dimension of the feature maps to convert them into probability distributions, and  $\phi_c'^{(l)}$  and  $\psi_S'^{(l)}$  are the local and global feature maps after passing through the alignment network.

### 3.3 Low-Overhead Network Strategy and Adaptive Aggregation

This framework addresses network challenges through an efficient network transmission strategy and a robust global aggregation mechanism. Directly transmitting high-dimensional feature maps  $\phi_c^{(l)}$  would consume a significant amount of the UAVs' precious uplink bandwidth. Therefore, we employ a FSC strategy, which utilizes adaptive average pooling to substantially downsample the feature map's spatial dimensions from  $H \times W$  to  $h \times w$ , generating  $\psi_c^{(l)}$ . For any given pixel  $(i, j)$  in the compressed feature map  $\psi_c^{(l)}$ , its computation is formulated as:

$$\psi_c^{(l)}(i, j) = \frac{1}{|R_{i,j}|} \sum_{(x,y) \in R_{i,j}} \phi_c^{(l)}(x, y) \quad (7)$$

where  $R_{i,j}$  is the adaptive pooling region in  $\phi_c^{(l)}$  that corresponds to  $(i, j)$ . The boundaries of this region  $R_{i,j}$  are dynamically determined by the input dimensions  $(H, W)$  and the output dimensions  $(h, w)$ .

This feature dimensionality reduction technique drastically reduces the data volume. Simultaneously, by using average pooling, it effectively filters out local noise while preserving the most critical structural information. Finally, the dimension-reduced feature map  $\psi_c^{(l)}$  is uploaded to the server along with the high-level semantic knowledge  $z_c$ .

To cope with the heterogeneity of data collected by different UAVs, the server employs adaptive weighted aggregation. The server first evaluates the data quality of each client  $c$  by defining three metrics. The specific definitions of these three metrics are as follows:

Class Coverage ( $M_c^{\text{cov}}$ ), which measures the proportion of classes held by client  $c$  relative to the total number of classes:

$$M_c^{\text{cov}} = \frac{|K_c|}{|K|} \quad (8)$$

where  $|K_c|$  is the size of the set of classes possessed by client  $c$ , and  $|K|$  is the total number of classes in the dataset.

Data Volume ( $M_c^{\text{vol}}$ ), which measures the total number of pixels held by client  $c$ , with logarithmic smoothing applied:

$$M_c^{\text{vol}} = \frac{\log(1 + N_c)}{N_{\max}} \quad (9)$$

where  $N_c$  is the total number of pixels on client  $c$ , and  $N_{\max}$  is a maximum pixel constant used for normalization.

Data Equilibrium ( $M_c^{\text{bal}}$ ), used to measure the balance of the local class distribution  $p_c$  on client  $c$ . We calculate this using the following formula:

$$M_c^{\text{bal}} = - \sum_{k \in K} p_c(k) \log(p_c(k)) \quad (10)$$

where  $K$  is the set of all classes, and  $p_c(k)$  is the probability of class  $k$  on client  $c$ . This formula calculates the information entropy of  $p_c$ ; a higher entropy value signifies a more balanced class distribution.

To approximate data quality and mitigate bias our aggregation strategy prioritizes clients with diverse data sufficient samples and balanced distributions. To achieve this dynamic weighting the server computes the average metrics of all clients in round  $t$  and normalizes them into dynamic weights  $[\alpha^{\text{cov}}, \alpha^{\text{vol}}, \alpha^{\text{bal}}]$ . We combine these metrics into the final dynamic contribution factor  $Q_c$ :

$$Q_c = \alpha^{\text{cov}} M_c^{\text{cov}} + \alpha^{\text{vol}} M_c^{\text{vol}} + \alpha^{\text{bal}} M_c^{\text{bal}} \quad (11)$$

During the aggregation phase, the server uses  $Q_c$  and the per-class pixel counts  $N_{c,k}$  as weights to compute the global logits  $z_{S,k}$  and global features  $\psi_{S,k}^{(l)}$ :

$$z_{S,k} = \frac{\sum_{c \in C_t} N_{c,k} \cdot Q_c \cdot z_{c,k}}{\sum_{j \in C_t} N_{j,k} \cdot Q_j} \quad (12)$$

$$\psi_{S,k}^{(l)} = \sum_{c \in C_t} \left( \frac{N_{c,k} \cdot Q_c}{\sum_{j \in C_t} N_{j,k} \cdot Q_j} \right) \psi_{c,k}^{(l)} \quad (13)$$

where  $z_{S,k}$  and  $\psi_{S,k}^{(l)}$  are the global logits for class  $k$  and global features at layer  $l$ , respectively, generated by the server's aggregation.  $C_t$  is the set of clients participating in aggregation in round  $t$ , with  $c$  and  $j$  serving as indices iterating through this set.  $z_{c,k}$  and  $\psi_{c,k}^{(l)}$  are the local logits and local features, respectively, uploaded by client  $c$ .

After the server completes the aggregation, it will broadcast these high-quality global logits  $z_{S,k}$  and global features  $\psi_{S,k}^{(l)}$  back to all participating clients as the global teacher knowledge. In the next training round, clients will use this global knowledge to guide the calculation of their distillation loss, thereby initiating a new round of local optimization.

## 4 Experiments and Analysis

### 4.1 Experimental Setup

To comprehensively evaluate the performance of our proposed CEFD framework, we conducted a rigorous comparison against a series of representative FL benchmarks. These benchmarks include the classic FedAvg, FedProx [15], the knowledge distillation-based FedDistill [10], and other advanced federated frameworks such as FedCross [9], FedBABU [21], and FedALA [28]. All experiments were performed on a unified software and hardware platform. We utilized the PyTorch 2.0.0 deep learning framework, with a hardware configuration consisting of an NVIDIA RTX 4090 GPU, an Intel Xeon Platinum 8481C CPU, and 90 GB of system RAM. For fairness of comparison all comparative methods employed DeepLabV3+ as the base segmentation network architecture. All experiments were trained using the SGD optimizer with an initial learning rate of 0.007 and run for a total of 150 rounds with local epochs set to 1 and a batch size of 4. To simulate UAV swarms of different

scales in post-flash-flood recovery networks, we established multiple FL scenarios comprising 4, 5, and 6 clients. The Landslide Segmentation Dataset (LSD) in this study is specifically designed for the semantic segmentation task of geological hazards, such as landslides and debris flows, in remote sensing imagery. The dataset is composed of high-resolution remote sensing satellite imagery, covering diverse geographical regions. The dataset provides precise pixel-level annotations, with labeled classes primarily including landslide area, debris flow area, and background area, making it an ideal benchmark for evaluating disaster monitoring algorithms.

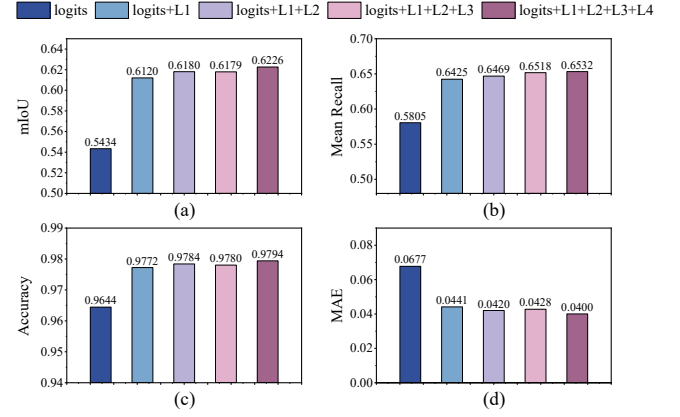
In the data preprocessing stage, we cropped all images from the LSD dataset into  $512 \times 512$  pixel patches to fit the input dimensions of the deep learning models. To enhance data diversity and improve the model's generalization capability, we applied data augmentation strategies to the training images, including random horizontal flipping, rotation, and color jittering. For the label data, nearest-neighbor interpolation was employed for synchronous processing to ensure the precision of the disaster area boundary annotations was preserved. To simulate the heterogeneity challenge of non-uniform data distribution among UAVs in real-world disaster scenarios, we adopted a Dirichlet distribution to partition the dataset. By setting the hyperparameter of the Dirichlet distribution ( $\alpha = 0.5$ ), we generated a class-imbalanced sample distribution for each client. This effectively simulates the real-world situation where different UAVs observe different types or quantities of disasters in different regions.

#### 4.2 Impact of the Number of Distillation Levels on Model Performance

We conducted a set of ablation experiments to validate the necessity of multi-level knowledge distillation within the CEFD framework and to determine the optimal number of distilled layers. We respectively distilled the following five configurations namely logits, logits+low-level features (logits+L1), logits+low-level features+aspp features (logits+L1+L2), logits+low-level features+aspp features+shortcut features (logits+L1+L2+L3), and logits+low-level features+aspp features+shortcut features+fused features (logits+L1+L2+L3+L4). All configurations were run under the same Non-IID data partition and experimental setup and were evaluated using four metrics mIoU, Mean Recall, Accuracy and MAE. The experimental results are illustrated in Fig. 2.

As clearly observed from the figure the number of distilled layers has a direct and significant impact on model performance. When only logits are distilled the model performance is at its minimum with an mIoU of only 0.5434 and an MAE as high as 0.0677. However once an additional intermediate feature layer is added logits+L1 all metrics show substantial improvement with mIoU increasing to 0.6120 and MAE decreasing to 0.0441. Building on this the performance gains from adding more distillation layers are more moderate but still exhibit a consistent positive trend. Ultimately the logits+L1+L2+L3+L4 configuration achieved the best performance across all four key metrics attaining the highest mIoU value of 0.6226 the highest Mean Recall of 0.6532 the highest Accuracy of 0.9794 and the lowest MAE value of 0.0400.

This result intuitively reflects that single-dimensional knowledge is insufficient to meet the demands of a complex segmentation



**Figure 2: Impact of the number of distillation levels on model performance. L1, L2, L3, and L4 are low-level features, aspp features, shortcut features, and fused features, respectively.**

task. The significant performance leap from Level Count=1 to Level Count=2, marked by an mIoU increase of 0.0686, demonstrates the strong complementarity between high-level semantics from logits and low-level spatial structure from feature layers. High-level knowledge alone cannot provide the model with the boundary and textural cues required for precise segmentation. As more levels were introduced, performance continued to climb, indicating that a richer combination of features spanning different levels of abstraction, ranging from shallow details to deep structures, is key to constructing robust global knowledge and guiding local model learning. Therefore, we selected the Level Count=5 configuration for all subsequent experiments.

#### 4.3 Impact of Pooling Strategy on Model Performance

We then analyzed the comprehensive impact of the FSC mechanism on model performance and network efficiency. To make the compression effect more concrete we specified the feature space dimensions before and after pooling. Based on our  $512 \times 512$  input and DeepLabV3+ architecture the original spatial dimensions ( $H \times W$ ) of the four layers of knowledge (L1-L4) are primarily  $128 \times 128$  (L1, L3, L4) and  $32 \times 32$  (L2). The four configurations we evaluated include CEFD transmitting the original dimensions, CEFD-1C which uniformly adaptively pools all feature maps to  $8 \times 8$  via one compression, CEFD-2C which pools to  $4 \times 4$  via two compressions, and CEFD-3C which pools to  $2 \times 2$  via three compressions. The experimental results are shown in Fig. 3.

In terms of model segmentation performance, as shown in the figure, CEFD-2C achieved the best performance across all four metrics. It reached an mIoU of 0.6226, a Mean Recall of 0.6532, an Accuracy of 0.9794, and the lowest MAE among all configurations at 0.0400. The other configurations, including the non-downsampled CEFD as well as CEFD-1C and CEFD-3C, exhibited similar performance levels, but none matched the performance of CEFD-2C.

This performance trend reflects a key trade-off in the feature dimensionality reduction process. Transmitting non-downsampled high-dimensional features or applying insufficient downsampling,

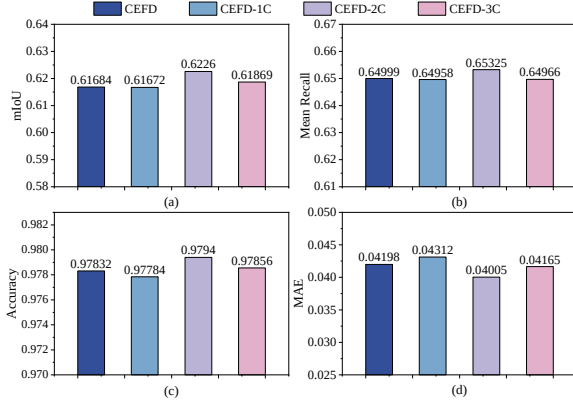


Figure 3: Impact of pooling strategy on model performance

such as in the CEFD and CEFD-1C configurations, retains excessive redundant information and local noise. This noise can contaminate the global knowledge during the aggregation process, thereby limiting the upper bound of model accuracy. Conversely, overly aggressive downsampling, as seen in the CEFD-3C configuration, while removing noise, also leads to the loss of critical spatial structure information, which similarly impairs segmentation performance. The CEFD-2C configuration performed best because it struck an ideal balance. While effectively filtering noise and distilling the most representative core knowledge, it also preserved the structural information necessary for precisely segmenting disaster boundaries. Therefore, we selected it as the standard configuration for the comparative experiments that followed.

#### 4.4 Performance Comparison of Different Federated Learning Algorithms under Non-IID Scenarios

To comprehensively and systematically evaluate the overall performance of our optimized CEFD framework in addressing the fundamental contradiction between accuracy and efficiency in post-flash-flood recovery networks, we conducted comparative experiments against a series of mainstream FL benchmarks. This comparative experiment aims to simultaneously quantify the segmentation accuracy of all methods under Non-IID data heterogeneity and the communication cost required for operation in UAV networks. Wherein, the communication cost for FedDistill and CEFD is defined as the memory occupied by the amount of knowledge exchanged, while the cost for other baseline algorithms is defined as the memory occupied by the amount of model parameters exchanged [10]. The experimental results are presented in Fig. 4 and Table 1, respectively.

As shown in the figure above, in the mIoU comparison, our method demonstrates a distinct advantage, achieving the highest mIoU and the fastest convergence speed across all client scales. The same trend is observed in the Accuracy comparison, where our method again establishes the optimal performance curve. FedALA achieves commendable results but still trails our method by a noticeable margin. In contrast, other methods, particularly FedDistill,

suffer from severely compromised segmentation accuracy, highlighting the immense challenge of collaborative training in a data-heterogeneous setting.

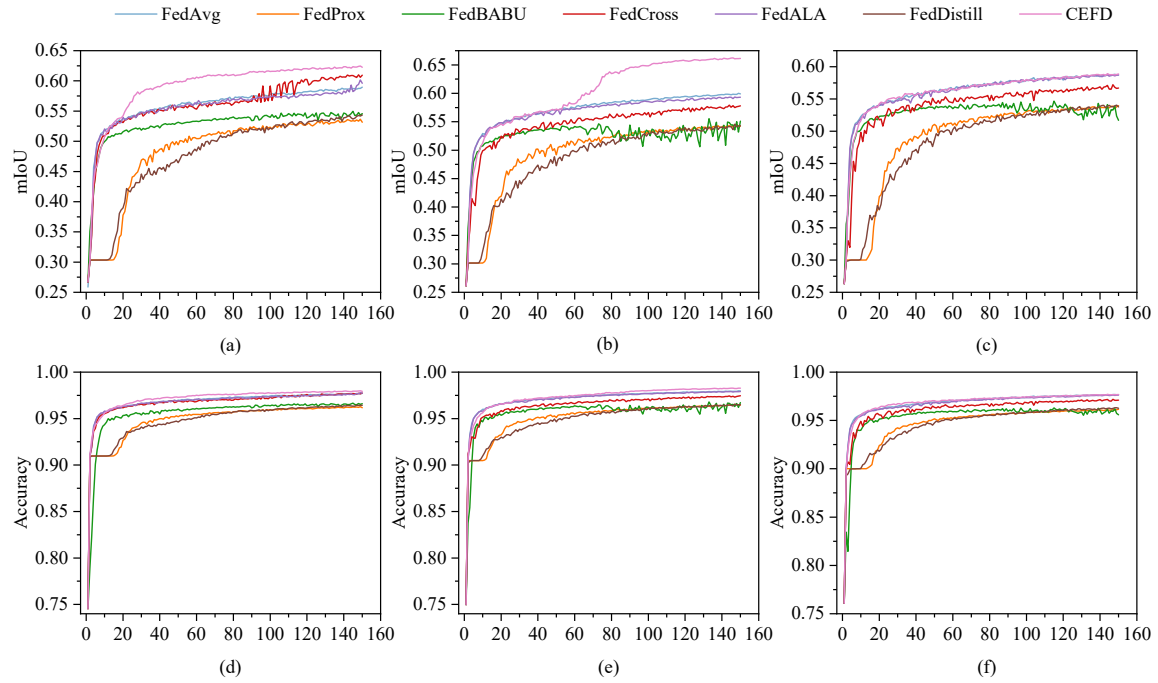
Network efficiency is the core bottleneck in disaster recovery networks. As shown in Table 1, which quantifies the communication cost for each method under different numbers of clients, a stark contrast is revealed. The vast majority of benchmark algorithms, such as FedAvg, FedProx, and FedALA, rely on parameter transmission, resulting in extremely high network overhead, reaching  $1.82 \times 10^5$  KB with just 4 clients. While FedDistill offers an extremely low cost (less than 1 KB), its accuracy, as shown in Fig. 4, is comparatively the worst among all methods. Current methodologies either attain requisite segmentation accuracy by imposing a substantial network burden, or they achieve minimal communication loads at the expense of severely compromised accuracy.

Table 1: Communication cost of different algorithms with varying numbers of clients

| Method          | Communication cost (KB) |                    |                    |
|-----------------|-------------------------|--------------------|--------------------|
|                 | NC=4                    | NC=5               | NC=6               |
| FedAvg [19]     | $1.82 \times 10^5$      | $2.27 \times 10^5$ | $2.73 \times 10^5$ |
| FedProx [15]    | $1.82 \times 10^5$      | $2.27 \times 10^5$ | $2.73 \times 10^5$ |
| FedDistill [10] | <b>0.4696</b>           | <b>0.5594</b>      | <b>0.6492</b>      |
| FedCross [9]    | $1.82 \times 10^5$      | $2.27 \times 10^5$ | $2.73 \times 10^5$ |
| FedBABU [21]    | $1.82 \times 10^5$      | $2.27 \times 10^5$ | $2.72 \times 10^5$ |
| FedALA [28]     | $1.82 \times 10^5$      | $2.27 \times 10^5$ | $2.73 \times 10^5$ |
| CEFD            | $8.76 \times 10^2$      | $1.06 \times 10^3$ | $1.24 \times 10^3$ |

CEFD method successfully resolves this dilemma. As shown in Table 1, the communication cost of CEFD at  $NC = 4$  is  $8.76 \times 10^2$  KB, approximately 99.5% lower than parameter aggregation methods such as FedAvg. For edge devices such as UAVs, communication energy consumption ( $E$ ) is closely related to the data transmission volume ( $D$ ) and transmission power ( $P$ ) ( $E \propto D \times P$ ). Assuming constant transmission power, the 99.5% communication overhead reduction achieved by CEFD translates directly into a nearly proportional reduction in communication energy consumption. This substantial enhancement in energy efficiency is paramount for battery-powered UAVs, as it directly extends the device's mission duration. While achieving this significant efficiency gain, the segmentation performance of CEFD is not compromised. On the contrary, it demonstrates state-of-the-art levels in Fig. 4, surpassing all benchmarks. This balanced superiority is a direct manifestation of its core architecture. The CEFD framework not only avoids the high network burden associated with transmitting bulky model parameters, as seen in methods like FedAvg, but it also overcomes the problem of incomplete information and severe accuracy degradation caused by relying solely on singular logits, as seen in methods like FedDistill.

CEFD achieves collaboration by exchanging a multi-level knowledge representation, which provides a form of complementary regularization. The high-level semantic logits offer contextual information about the classes, while the critical intermediate-layer spatial structure information provides pixel-level spatial evidence. This



**Figure 4: Performance comparison of different algorithms under different numbers of clients. The top row shows the mIoU comparison with (a) NC=4, (b) NC=5, and (c) NC=6. The bottom row shows the Accuracy comparison with (d) NC=4, (e) NC=5, and (f) NC=6.**

combination of knowledge ensures the global model possesses both correct semantic guidance and precise boundary awareness. Looking deeper, the feature spatial dimensionality reduction plays a dual role here. It serves as both a dimensionality reduction tool for network efficiency and as an active information purification mechanism. Through adaptive pooling, this process effectively filters out high-frequency, sample-specific noise and redundant details from the client's local feature maps. This compels clients to share only core, generalizable structural information, naturally filtering out local noise to yield a lighter, robust global model. Such distilled knowledge representation underpins CEFD's efficiency in resource-constrained networks and its capability to overcome Non-IID heterogeneity for high-precision segmentation.

## 5 Conclusion

This study aims to resolve the conflict between high-precision mapping and constrained UAV communication in flash flood disaster assessment, a scenario where existing Federated Learning approaches struggle to balance communication overhead with accuracy. To this end, we propose the CEFD framework, which features a lightweight multi-level knowledge representation. Specifically, by forgoing model parameter transmission in favor of exchanging high-level semantics and compressed spatial features, this mechanism significantly reduces network load while effectively preserving critical information required for precise segmentation. Experimental results show that CEFD achieved high segmentation performance on the LSD dataset, attaining an mIoU of 0.6226, a Mean Recall of

0.6532, an Accuracy of 0.9794, and an MAE of 0.0400. Simultaneously, its communication cost was reduced by approximately 99.5% compared to methods like FedAvg, successfully achieving a balance between accuracy and efficiency.

Although this work focuses on the core algorithm CEFD's significantly reduced payload makes it inherently more resilient to real-world network challenges as the probability of successful transmission is inversely proportional to packet size. Formal validation under simulated network constraints is a key direction for future work.

## 6 Acknowledgment

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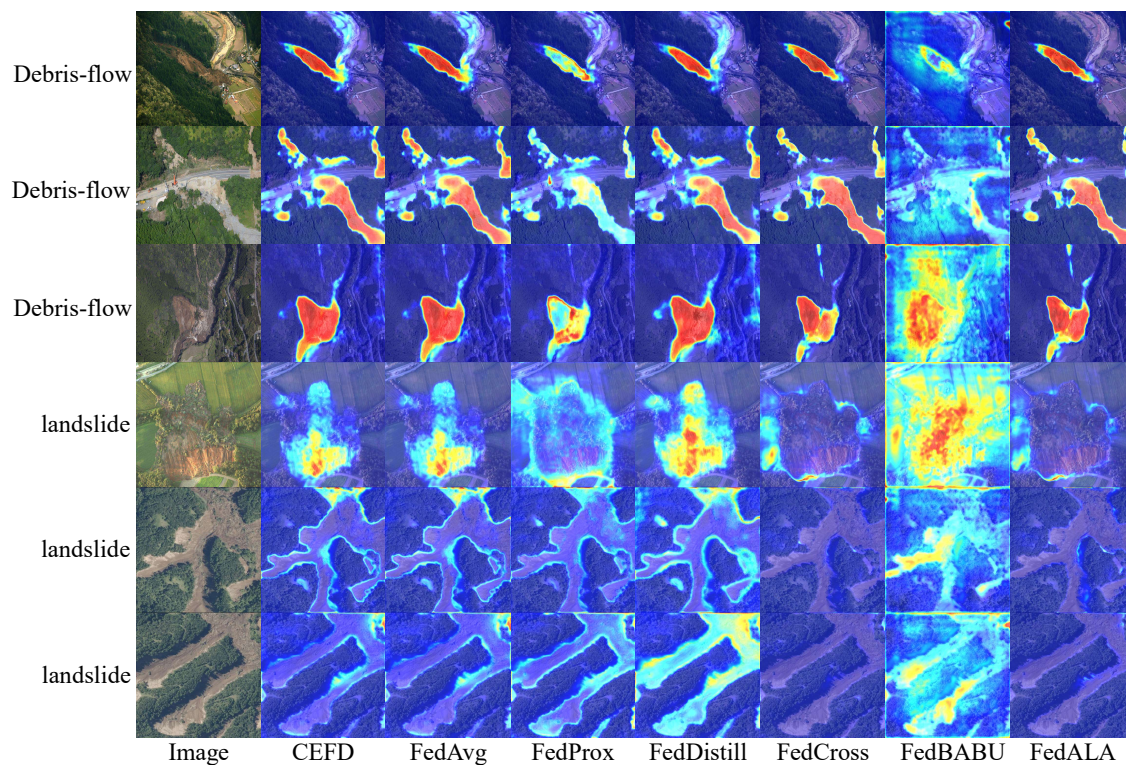
## A APPENDIX

### A.1 Qualitative Comparison and Analysis of Segmentation Results from Different Algorithms

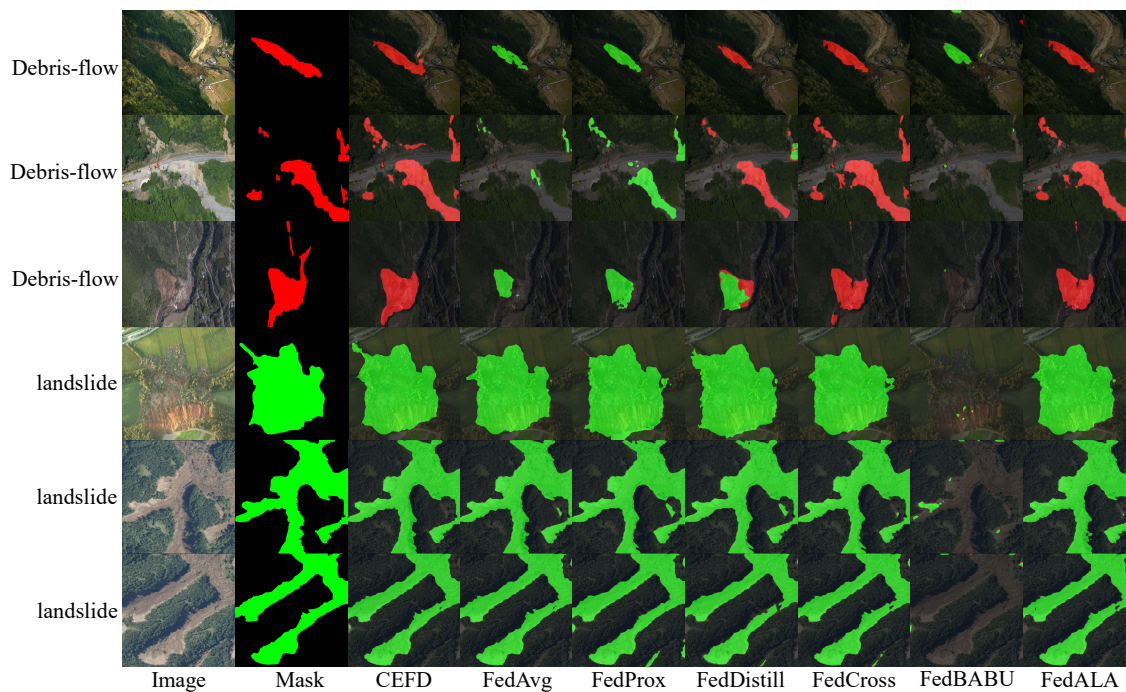
To more intuitively compare the practical performance of different algorithms on the disaster assessment segmentation task, we conducted further qualitative visualizations. We provide two types of visual evidence. The first consists of confidence heatmaps, which illustrate the confidence level of each method when predicting target classes. Warm colors represent high model confidence that a pixel belongs to a disaster class, while cool colors represent low confidence. The second is the final predicted segmentation mask, which visually presents the model's output. In these masks, red represents debris flow, and green represents landslide. All these visual comparisons are shown in Fig. a1 and Fig. a2.

The prediction maps in Fig. a2 and the heatmaps in Fig. a1 visually demonstrate the significant advantages of the CEFD framework in terms of segmentation accuracy and feature-learning robustness. As shown in Fig. a2, the prediction results of our CEFD method are highly consistent with the ground truth across all disaster scenarios. Whether it is the narrow, elongated morphology of a debris flow or the irregular, complex boundaries of a landslide, CEFD achieves precise and complete delineation with sharp edges. In contrast, the other methods all perform poorly. FedDistill, which relies solely on logit distillation, produces prediction maps that, while roughly accurate in location, are highly fragmented and incomplete, falling far short of the needs for disaster assessment. This visually confirms the hypothesis that high-level semantic logits alone cannot transfer the low-level spatial structure information necessary for the segmentation task.

The methods relying on parameter aggregation all performed poorly in the test scenarios. As shown in Fig. a2, the segmentation results from FedAvg and FedProx exhibit severe class confusion. In the debris flow scenario, they consistently misclassify the disaster area that should be debris flow as a landslide. In the landslide scenario, although they identify the correct class, they fail to reproduce the complex boundaries and details shown in the Ground Truth, leading to incomplete segmentation and shape distortion. FedBABU's performance was the worst; its segmentation results are almost empty. Its heatmaps in multiple scenarios show large areas of diffuse high-level activation, completely failing to focus on the disaster targets.



**Figure a1: Heatmaps of different algorithms.** The first three rows show the heatmap analysis for debris flow samples, and the last three rows show the heatmap analysis for landslide samples.



**Figure a2: Prediction maps of different algorithms.** The first three rows show the segmentation results for debris flow samples, and the last three rows show the segmentation results for landslide samples.

The precise focusing capability and high segmentation accuracy demonstrated by CEFD in Fig. a1 and Fig. a2 intuitively reflect the core innovations of its framework. CEFD discards the bulky parameter transmission of FedAvg and the singular logit transmission of FedDistill. Instead, it designs an information-rich yet lightweight knowledge representation. This representation combines two complementary types of information: it contains both the high-level semantics that define the disaster class and the key spatial evidence—the mid-level features—that locate the disaster's boundaries. The latter encapsulates the texture and structural details necessary for precise disaster delineation. To efficiently transmit these mid-level features, CEFD employs a feature spatial compression strategy. This strategy not only reduces the data volume through substantial dimensionality reduction but, more importantly, it acts as an active information purification mechanism. By using average pooling, this method effectively filters out high-frequency, sample-specific noise and redundant details from the client's local feature maps. This mechanism ensures that clients share only the most generalizable structural information, making the global knowledge aggregated by the server purer and thus enabling excellent segmentation results even in complex scenarios.

## A.2 Derivation of the Dynamic Contribution Factor $Q_c$

In Eq. 11, we introduced the dynamic contribution factor  $Q_c$ , which is determined by three adaptive coefficients:  $\alpha^{cov}$ ,  $\alpha^{vol}$ , and  $\alpha^{bal}$ . This appendix details the calculation process for these coefficients. This calculation process is divided into two steps:

First, the server aggregates the three quality metrics from all clients participating in the aggregation in round  $t$  (the set  $C_t$ ). The server then calculates the average value of these three metrics across all participating clients to obtain the average metric vector  $\bar{M}_t = [\bar{M}^{cov}, \bar{M}^{vol}, \bar{M}^{bal}]$ :

$$\bar{M}^{cov} = \frac{1}{|C_t|} \sum_{c \in C_t} M_c^{cov} \quad (a1)$$

$$\bar{M}^{vol} = \frac{1}{|C_t|} \sum_{c \in C_t} M_c^{vol} \quad (a2)$$

$$\bar{M}^{bal} = \frac{1}{|C_t|} \sum_{c \in C_t} M_c^{bal} \quad (a3)$$

Next, the server normalizes these three average metrics to compute their relative contribution to the final weight. First, the sum of the average metrics,  $S_{total}$ , is calculated:

$$S_{total} = \bar{M}^{cov} + \bar{M}^{vol} + \bar{M}^{bal} \quad (a4)$$

Then, by dividing each average metric by the sum, the dynamic weight coefficients for the current round,  $\alpha^{cov}$ ,  $\alpha^{vol}$ , and  $\alpha^{bal}$ , are obtained:

$$\alpha^{cov} = \frac{\bar{M}^{cov}}{S_{total}} \quad (a5)$$

$$\alpha^{vol} = \frac{\bar{M}^{vol}}{S_{total}} \quad (a6)$$

$$\alpha^{bal} = \frac{\bar{M}^{bal}}{S_{total}} \quad (a7)$$

These dynamic coefficients are subsequently used to calculate the final contribution factor  $Q_c$  for each client  $c$ , as shown in Eq. 11 in the main text.

## A.3 Dataset and code link

Data are openly accessible at [https://github.com/zhaoyk926/landslide\\_segmentation-LSD-.git](https://github.com/zhaoyk926/landslide_segmentation-LSD-.git).

The code is publicly available at <https://github.com/zhaoyk926/CEFD.git>.